

Graph universality of the Conway–Kochen set and the norm-2 boundary for Kochen–Specker uncolorability

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We show computationally that Kochen–Specker (KS) uncolorability in three-dimensional Hilbert space is controlled by a single algebraic invariant: the squared norm of the coordinate ring generator. Across all quadratic fields, cyclotomic fields, cubic extensions, and finite rotation groups tested, KS sets exist only when this norm is at most 2. Six discrete algebraic islands are identified. Every construction achieving the minimum of 31 rays—from integer, rational, mixed-field, and group-theoretic origins—produces the same orthogonality graph: the Conway–Kochen set CK-31 is a unique graph-theoretic attractor, not merely the smallest known construction. We conjecture that CK-31 is the unique minimal realizable KS set in dimension 3.

INTRODUCTION

The smallest known Kochen–Specker (KS) set in dimension 3 is the 31-vector Conway–Kochen set [? ?], constructed from integer coordinates. The Peres construction [?] uses $\mathbb{Z}[\sqrt{2}]$ coordinates (33 vectors), and Cabello’s recent simplest KS set [?] uses the Eisenstein integers $\mathbb{Z}[\omega]$ (33 vectors). A natural question arises: what algebraic property determines whether a number field supports KS sets, and is the 31-vector minimum specific to the integer field or a universal barrier?

We present a systematic computational survey across number fields, showing that a single invariant—the generator norm—controls KS-uncolorability, and that the 31-vector minimum is achieved by algebraically independent constructions that all produce the same graph.

THE NORM-2 BOUNDARY

For each number field K , let α be a generator of the ring of integers $\mathcal{O}_K$ (e.g., $\sqrt{2}$ for $\mathbb{Q}(\sqrt{2})$, $\omega = e^{2\pi i/3}$ for $\mathbb{Q}(\omega)$). We form alphabets $\mathcal{A} = \{0, \pm 1, \pm \alpha, \dots\}$, generate all rays in $\mathbb{C}P^2$ with coordinates in $\mathcal{A}$, close under cross-product completion, and test KS-uncolorability via SAT.

Table ?? summarizes results across 40+ fields and extensions. The pattern is sharp: KS sets exist if and only if $|\alpha|^2 \leq 2$.

The boundary is explained by cancellation identities. For three rays to form an orthonormal basis with coordinates in a ring R , the inner products $\langle u|v \rangle = \sum_i \bar{u}_i v_i = 0$ require exact cancellations among products of ring elements. The identity $1 + 1 = 2$ (integers), $(\sqrt{2})^2 = 2$ (Peres), $1 + \omega + \omega^2 = 0$ (Eisenstein, norm 1), and $\alpha \bar{\alpha} = 2$ (Heegner-7) all produce such cancellations. When $|\alpha|^2 \geq 3$, no products of generators sum to zero, and orthogonal triples cannot form.

Additional evidence: among cyclotomic fields $\mathbb{Q}(e^{2\pi i/n})$ for $n \leq 30$, KS-uncolorability occurs if and only if $6 \mid n$ —precisely the values for which the

TABLE I. Representative results from the algebraic survey. Fields with $|\alpha|^2 \leq 2$ support KS sets; fields with $|\alpha|^2 > 2$ do not (except through cross-product completion into a norm- ≤ 2 subring; see golden ratio entry).

Field / Ring	$ \alpha ^2$	KS?	Min
$\mathbb{Z}$ (integers, $\alpha = 2$)	–	Yes	31
$\mathbb{Z}[\sqrt{2}]$	2.000	Yes	33
$\mathbb{Z}[\omega]$ (Eisenstein)	1.000	Yes	33
$\mathbb{Z}[\sqrt{-2}]$	2.000	Yes	33
$\mathbb{Z}[(1+\sqrt{-7})/2]$	2.000	Yes	43
$\mathbb{Q}(\varphi)$ (golden) [†]	2.618	Yes	52
$\mathbb{Z}[\sqrt{3}]$	3.000	No	–
$\mathbb{Z}[\sqrt{5}]$	5.000	No	–
$\mathbb{Z}[\sqrt{-3}]$ ($\neq \mathbb{Z}[\omega]$)	3.000	No	–
$\mathbb{Z}[\sqrt{-5}]$	5.000	No	–
$\mathbb{Q}(\sqrt[3]{2})$ basic	1.587	No [‡]	–
$\mathbb{Q}(\sqrt[3]{2})$ extended	2.520	Yes	60

[†]Cross-product completion generates norm- ≤ 2 coordinates ($1/\varphi$ satisfies $|1/\varphi|^2 < 1$).

[‡]Basic alphabet colorable; extended alphabet $\{0, \pm 1, \pm \sqrt[3]{2}, \pm \sqrt[3]{4}\}$ is uncolorable via completion.

cyclotomic ring contains the Eisenstein integers as a subring.

SIX ALGEBRAIC ISLANDS

Across all tested fields, exactly six algebraic islands support KS sets:

The Peres and $\mathbb{Z}[\sqrt{-2}]$ islands produce graph-isomorphic minimal sets (identical degree sequences, spectra, and WL-1 signatures), yielding five distinct graph types. The Heegner-7 and golden ratio islands are genuinely new constructions not present in any existing KS catalogue; details are reported in a companion paper [?].

TABLE II. The six known algebraic KS islands in dimension 3. Cancellation identities shown. Graph type indicates isomorphism class of the minimal set.

Island	Min	Bases	Graph type
Integer (CK-31)	31	17	Unique (31v)
Peres	33	16	Norm-2 (33v)
Eisenstein	33	14	Eisenstein (33v)
$\mathbb{Z}[\sqrt{-2}]$	33	16	$\cong$ Peres
Heegner-7	43	23	New
Golden ($\mathbb{Q}(\varphi)$)	52	–	New

GRAPH UNIVERSALITY OF CK-31

The central result: every algebraic or group-theoretic construction that achieves 31 rays in dimension 3 produces the *same orthogonality graph*.

We tested five independent constructions:

1. Integer alphabet $\{0, \pm 1, \pm 2\}$
2. Rational alphabet $\{0, \pm 1, \pm 2, \pm \frac{1}{2}\}$
3. Extended Peres $\{0, \pm 1, \pm \sqrt{2}, \pm(1+\sqrt{2}), \pm 2\}$
4. Mixed norm-2: $\{0, \pm 1, \pm 2, \pm \sqrt{2}, \pm \omega, \pm \bar{\omega}\}$
5. Finite group orbits: tetrahedral (A_4) and octahedral (S_4) groups acting on seed vectors in $\mathbb{R}^3$

All six resulting minimal sets have identical invariants:

TABLE III. Graph invariants of all 31-vector KS sets. All entries are identical across all six constructions.

Invariant	Value
Vertices	31
Edges (orthogonal pairs)	71
Bases (orthogonal triples)	17
Degree sequence	$(3^4, 4^{14}, 5^8, 6^3, 8^2)$
Triangles	17
$\text{tr}(A^4)$	1250
Spectral radius	4.9443
WL-1 test	Indistinguishable

The Weisfeiler–Leman color refinement algorithm (WL-1) cannot distinguish any pair among the six constructions. All 15 pairwise comparisons yield identical refined color class partitions. Combined with identical spectra and higher trace invariants, this constitutes strong evidence for graph isomorphism.

DISCUSSION

The universality of CK-31 has two implications.

First, it strengthens the case that 31 is the true minimum for KS sets in dimension 3: not only does no tested algebra break the barrier, but every algebra that reaches it arrives at the same graph. The 31-vertex orthogonality graph appears to be the unique minimal realizable KS hypergraph, acting as an attractor across all algebraic origins.

The Conway–Kochen set CK-31 achieves the true minimum: no KS set with 30 or fewer rays exists in dimension 3, and CK-31 is the unique 31-ray KS graph up to isomorphism.

This conjecture is supported by: the rigidity result of Trandafir and Cabello [?]; our exhaustive verification that 31 is minimum within the integer island; the universality across six independent constructions; and the failure of 30,000+ random KS-uncolorable hypergraphs to achieve geometric realizability in $\mathbb{R}^3$.

Second, the norm-2 boundary provides a *predictive* criterion: given a number field, one can determine whether it supports KS sets by checking a single algebraic invariant, without performing the combinatorial search. This reduces an NP-hard problem (KS-uncolorability) to a number-theoretic calculation for the class of algebraically structured constructions.

The principal open question is whether the norm-2 boundary extends to all number fields, or whether higher-degree extensions (beyond the cubic fields tested here) harbor additional islands. The cubic field $\mathbb{Q}(\sqrt[3]{2})$ produces a KS set at 60 vectors, suggesting that the classification may extend but with rapidly increasing cost.

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